SAFI'S LAB

Ender-3 NG | Bill of Materials

From-scratch build - repository BOM edition - 22 October 2026

Source: safi-toumi/Ender-3-NG-from-scratch, BOM.md. Quantities, alternatives and specifications below follow the repository. This document is a formatted transcription, not an independently validated build specification.

Supplier note: the source specifies V-slot extrusions. UIF Equipment offers T-slot profiles; verify cross-sections, printed-part fit, slot dimensions and compatible hardware before substituting.

Frame & Enclosure

Part	Qty	Notes
4040 V-slot extrusion, 350 mm	1	Base frame rear
4040 V-slot extrusion, 290 mm	2	Base frame sides/ M5 taped end holes
2040 V-slot extrusion, 400 mm	2	Frame vertical supports / M5 taped end holes
2040 V-slot extrusion, 350 mm	2	Ultimate frame front (fit 310 mm space with printed spacers)
2040 V-slot extrusion, 310 mm	2	
2020 V-slot extrusion, 330 mm	1	Top cross frame (replaces original Ender 3 Y extrusion)
2020 V-slot extrusion, 345 mm	1	M5 taped end holes
Rubber feet (19 mm, M5 hole)	4	ensure good quality and vibration handling
4 mm polycarbonate panel	3-4	Sides & top for V2-style enclosure
5 mm polycarbonate panel	1	Front door
3 mm panel	1	Rear electronics cover
M5 T-nuts & M5x10/12 BHCS	~40	For joining extrusions (quantities vary with frame)

For more info on the polycarbonate pannel and the fastners visit: rh3d.xyz

Motion System

Part	Qty	Notes
NEMA 17 stepper motor (2.5A)	2	For CoreXY gantry (good quality ones will make a diff)
NEMA 17 stepper motor (standard)	2-3	For Z-axis & extruder (depending on single/triple Z)
8 mm linear rod, 350 mm	4	X/Y gantry
12 mm linear rod, 350 mm	2	Z-axis (main)
12 mm linear rod, 320 mm	1	Z-axis (triple Z option)
LM8LUU linear bearing	4	ensure good quality!
LM12UU linear bearing	3	ensure good quality!
GT2 6 mm closed-loop belt, 1140 mm	1	Z-axis belt
GT2 6 mm open belt (Gates)	3 m	For CoreXY AB belts
40T GT2 pulley (8 mm bore, 6 mm belt)	3	
20T GT2 pulley (5 mm bore, 6 mm belt)	3	
F695-2RS bearing	32	For gantry & belts
608-2RS bearing	6	
T8/8 x 300 mm leadscrew	3	For triple Z (or 1 for single Z)
T8/8 leadscrew nut	3	

Electronics & Toolhead

Part	Qty	Notes
32-bit mainboard	1	BTT SKR Mini E3 / Octopus etc. (Klipper recommended)
24 V power supply	1	350 W or higher recommended
Hotend (e.g., Volcano, Dragon, etc.)	1	Your choice; matches your printed toolhead
Extruder (e.g., BMG, Orbiter)	1	Direct drive recommended
3010 axial fan, 24 V	1	Hotend cooling
4010 radial fan, 24 V	2	Part cooling
6015 axial fan, 24 V	2-3	Electronics & enclosure
DIN rail, 350 mm	2	For electronics enclosure
DIN rail, 200 mm	1	
Microswitch (Omron D2F-L)	3	Endstops
PTFE tube 4/2 mm, 1 m	1	
Bed silicone spacers	4	
Heated bed	1	Reuse or buy new (e.g., 235x235 mm)
PEI sheet / build surface	1	

Hardware & Fasteners (Typical Quantities)

This is a consolidated hardware list from the HoneyBadger kit. Quantities may vary slightly depending on your exact printed parts, so treat this as a shopping checklist and verify with the Configurator.

Type	Sizes & Quantities
M2 SHCS	M2x10 - 4 pcs
M3 SHCS	M3x6 - 72, M3x8 - 50, M3x10 - 34, M3x16 - 4, M3x20 - 15, M3x25 - 8, M3x35 - 11
M4 SHCS	M4x8 - 34, M4x10 - 12
M5 SHCS	M5x12 - 24, M5x20 - 4, M5x25 - 10, M5x35 - 4
M5 BHCS	M5x10 - 4, M5x20 - 20, M5x30 - 4
M3 set screws	M3x5 flatpoint - 8
Washers	M5 (5.3x10x1 mm) - 65
Nuts	M3 - 5
T-nuts	M3 for 2020 - 35, M4 for 2020 - 32
Heat inserts	M3 (4x4.5 mm) - 110, M5 (5x7 mm) - 24

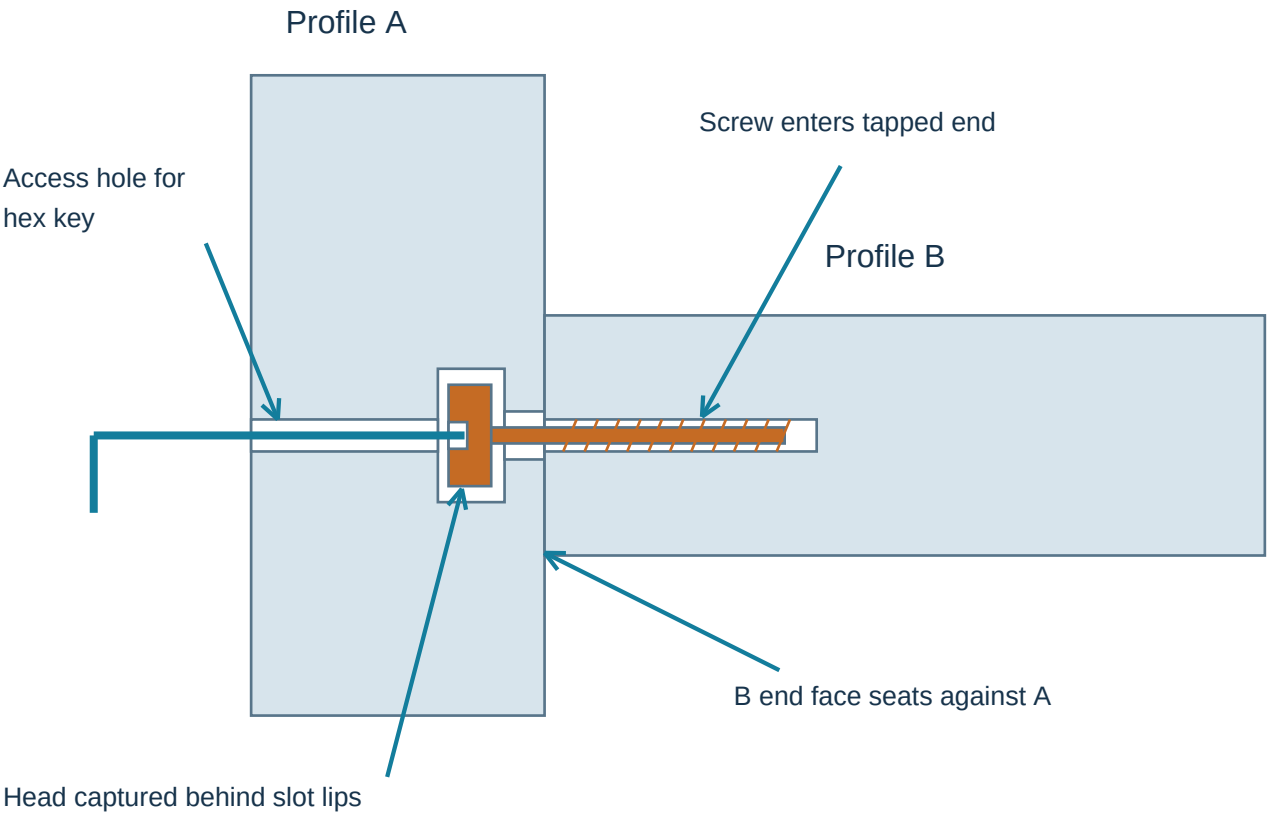
Printed Parts

You'll need to print a substantial number of parts from the official STL files on Printables. The Configurator will generate a printed parts list tailored to your configuration. Make sure to print with a well-calibrated printer and the recommended filament (e.g., ABS/ASA/PETG for structural parts).

Blind joints: how they work

A concealed end-screw joint connects the tapped end of profile B to the side slot of profile A. The screw head is captured inside A; an access hole lets a hex key reach its socket.

1 ASSEMBLED JOINT - CUTAWAY



Orange: screw. Blue: hex key. White spaces show the tool path and fastener clearance. The internal webs of the real extrusion are omitted for clarity.

WHAT MAKES THE JOINT CLAMP?

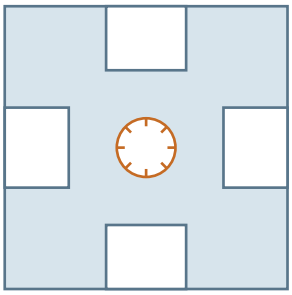
Tightening pulls B toward A while the screw head bears against the inner slot shoulders. A compatible head or the specified end-fastener clip must engage both shoulders; it must not slip through the slot mouth.

BEFORE USING UIF T-SLOT PROFILES

Check the actual profile drawing and test one sample. Slot-mouth width, internal clearance, core bore and screw-head geometry determine compatibility. Do not assume that any M5 screw will work simply because the profile is 2020 or 2040.

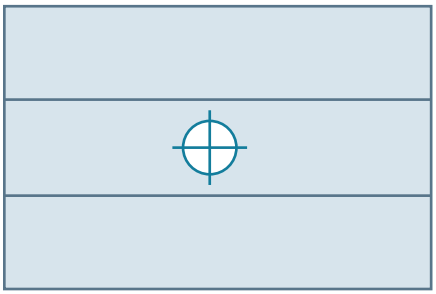
Preparation and assembly

2 PREPARE THE TWO PROFILES



B: end view

Tap the core bore only if its size and wall thickness suit the chosen thread.



A: tool-access face

Locate the access hole on the screw axis at the final joint position.

Schematic end views only. Hole diameter, tapping size, bolt length and tightening torque must come from the actual profile and fastener specification. Deburr all drilled holes.

3 ASSEMBLE IN THIS ORDER

1. Start the screw in B and leave enough clearance under its head for the slot shoulders.
2. Slide the head or specified end-fastener assembly into A from an open slot end.
3. Move B to the final position; align the socket with the access hole in A.
4. Hold the profiles square, tighten through the access hole, then recheck squareness.

FRAME DETAILS

A single concealed screw can permit rotation if the joint is not adequately restrained. Use the specified anti-rotation feature, a second fastener where the profile allows it, or suitable brackets/plates where the frame design requires them. Keep slot ends accessible during assembly.

Reference: 80/20 Standard End Fastener, M5 (20-3895): end tap and access hole; its assembly includes a wing clip. This drawing explains the principle, not an interchangeable part specification.

https://8020.net/20-3895.html